

Vibrations

$$u(t) = u_c(t) + u_p(t) = \text{total response}$$

$$m\ddot{u} + c\dot{u} + ku = F(t) \implies \ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = \frac{F(t)}{m}$$

$$\zeta = \frac{c}{2m\omega_n} \quad \omega_n = \sqrt{\frac{k}{m}} \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

Standard Form Conversions

$$u(t) = \begin{cases} 1) & B \cos(\omega t) + D \sin(\omega t) \\ 2) & A \cos(\omega t + \phi) \\ 3) & C_1 e^{j\omega t} + C_1^* e^{-j\omega t} \\ 4) & \text{Re}\{C e^{j\omega t}\} \end{cases}$$

1) - 2)	$B = A \cos(\phi) \quad D = -A \sin(\phi)$ $A = \sqrt{B^2 + D^2} \quad \phi = \text{atan2}(-D, B)$
1) - 3)	$B = 2\text{Re}\{C_1\} \quad D = -2\text{Im}\{C_1\}$ $C_1 = \frac{1}{2}B - \frac{1}{2}jD$
1) - 4)	$B = \text{Re}\{C\} \quad D = -\text{Im}\{C\}$ $C = B - jD$
2) - 3)	$A = \sqrt{2} C_1 \quad \phi = \text{atan2}(\text{Im}\{C_1\}, \text{Re}\{C_1\})$ $C_1 = \frac{A}{\sqrt{2}} e^{j\phi}$
2) - 4)	$A = C \quad \phi = \text{atan2}(\text{Im}\{C\}, \text{Re}\{C\})$ $C = A e^{j\phi}$
3) - 4)	$C = 2C_1 \quad C_1 = C/2$

Solutions

Standard 2nd order:

$$u_c(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \quad u_p(t) = \text{matches order of forcing}$$

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} = -\zeta\omega_n \pm j\omega_d \in \mathbb{C}$$

Free and Undamped:

$$u(t) = \begin{cases} C e^{j\omega_n t} + C^* e^{-j\omega_n t} \\ u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) \end{cases} \quad \lambda_{1,2} \in \mathbb{C} \setminus \mathbb{R}$$

Underdamped:

$$u_p(t) = e^{-\zeta\omega_n t} (a \cos(\omega_d t) + b \sin(\omega_d t)) = e^{-\zeta\omega_n t} \cos(\omega_d t + \phi)$$

$$T_d = \frac{2\pi}{\omega_d} \quad \left(\frac{\omega_d}{\omega_n}\right)^2 + \zeta^2 = 1 \quad \lambda_{1,2} \in \mathbb{C}$$

Overdamped:

$$u_p(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \quad \lambda_{1,2} \in \mathbb{R}$$

$$\begin{cases} u(t=0) = u_0 = C_1 + C_2 \\ \dot{u}(t=0) = v_0 = \lambda_1 C_1 + \lambda_2 C_2 \end{cases}$$

Critically Damped:

$$u_p(t) = C_1 e^{\lambda t} + C_2 t e^{\lambda t} \quad \lambda_{1,2} \in \mathbb{R} \quad t \because \text{ic's bad otherwise}$$

Harmonic Excitation:

$$u_p(t) = A(\omega) \cos(\omega t + \phi(\omega)) \quad A(\omega) \text{ implies } A(\omega) \in \mathbb{R}_+$$

$$\text{FRF} \triangleq \frac{A(\omega)}{F_0} \quad \Omega \triangleq \frac{\omega}{\omega_n} \begin{cases} \in (0, 1) & \phi(\Omega) = 0 \\ = 1 & \text{Resonance} \\ > 1 & \phi(\Omega) = \pi \end{cases}$$

$$u_p(t, \Omega) = \frac{F_0}{m\omega_n^2} \frac{1}{1 - \Omega^2} \cos(\omega t)$$

Note: Resonance grows linearly with time.

Equation of Motion

- Newton's Law - FBD - $\sum F = m\ddot{u}$
- Euler's Law - FBD - $\sum M = J\ddot{\theta}$
- Energy Conservation $KE + PE = \text{const}$ (dissipation not covered)
- Lagrangian $L = KE - PE$, $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad \forall q$ (dissipation not covered)

Fourier Series

$$f(t) = \sum_{n=0}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t) \quad T = 2\pi\omega$$

$$a_n = \frac{\omega}{\pi} \int_0^T f(t) \cos(n\omega t) dt \quad b_n = \frac{\omega}{\pi} \int_0^T f(t) \sin(n\omega t) dt$$

a_n and b_n are inner products with each of the sinusoidal bases

General Excitation

$$f(t) = \begin{cases} 1) \text{ Polynomial} \\ 2) \text{ Harmonic} \\ 3) \text{ Periodic} \rightarrow \text{Fourier Series} \\ 4) \text{ Impulse} \\ 5) \text{ Aperiodic} \rightarrow \text{Convolve} \end{cases}$$

Impulse: equivalent forms

$$m\ddot{h} + c\dot{h} + kh = \delta(t), \quad h(0) = 0, \quad \dot{h}(0) = 0$$

$$m\ddot{h} + c\dot{h} + kh = 0, \quad h(0) = 0, \quad \dot{h}(0) = 1/m$$

$$\text{if } \zeta \in (0, 1) \quad h(t) = \frac{1}{m\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t)$$

Convolution:

$$u_p(t) = g(t) * h(t) = \int_0^t g(\tau) h(t - \tau) d\tau \quad g(t) = \text{any}$$

MDOF

Eigenfrequencies: $\{\omega_i\}$ Eigenvalues: $\{\omega_i^2\}$
Mode shapes: Mass-normalized eigenvectors form an orthonormal basis (in modal space) of all possible vibrations

$$\vec{u}_i = \alpha_i [u_{i1} \quad \dots \quad u_{in}]^T \quad \vec{u}_i^T M \vec{u}_i = 1 \quad \vec{u}_i^T K \vec{u}_i = \omega_i^2$$

Undamped

1. Solve transient/complementary $\vec{x} = \vec{u}e^{j\omega t}$ $\vec{u} \in \mathbb{C}^n$

$$M\ddot{\vec{x}} + K\vec{x} = \vec{0} \quad \text{char eq: } \det(K - \omega^2 M) = 0$$

$$\text{eigenvalues: } \{\omega_i^2\} \quad \text{eigenvectors: } \{\vec{u}_i\}$$

$$\text{Modal matrix: } U = [\vec{u}_1 \ \dots \ \vec{u}_n]$$

$$[\vec{u}_1 \ \dots \ \vec{u}_n], \text{diag}(\lambda_1 \ \dots \ \lambda_n) = \text{eig}(K, M)$$

2. Diagonalize into decoupled ODEs of modal coordinates

$$\vec{x} = U\vec{q} \quad S = \text{diag}(\omega_1^2 \ \dots \ \omega_n^2)$$

$$U^T M U \ddot{\vec{x}} + U^T K U \vec{x} = U^T \vec{f} \sin(\omega t) \iff I \ddot{\vec{q}} + S \vec{q} = \vec{g} \sin(\omega t)$$

Excitation of $\vec{f}$ to mode i is $|\vec{u}_i^T \vec{f}|$

3. Independently solve each SDOF modal ODE

$$\ddot{q}_i + \omega_i^2 q_i = g_i \sin(\omega t) \quad q_i(t) = q_{ic}(t) + q_{ip}(t)$$

4. Inverse transform to physical coordinates, and plug ics into **total solution** to solve for remaining undetermined complementary solution coefficients $\vec{x} = U\vec{q}$

Proportional Damping

Let/assert: $C = \alpha M + \beta K$ so that modal analysis can be used to find natural frequencies and mode shapes

1. Ignore C , find eigenvalues and eigenvectors

$$\text{eigenvalues: } \{\omega_i^2\} \quad \text{eigenvectors: } \{\vec{u}_i\}$$

2. Diagonalize into decoupled ODEs of modal coordinates using U^T

$$M\ddot{\vec{q}} + C\dot{\vec{q}} + K\vec{q} = \vec{f} \sin(\omega t)$$

$$I\ddot{\vec{q}} + (\alpha I + \beta S)\dot{\vec{q}} + S\vec{q} = U^T \vec{f} \sin(\omega t)$$

3. Compare to standard form to find damping

$$\text{compare: } \begin{cases} \ddot{q}_i + (\alpha + \beta\omega_i^2)\dot{q}_i + \omega_i^2 q_i = (\vec{u}_i^T \vec{f}) \sin(\omega t) \\ \ddot{q}_i + (2\zeta_i\omega_i)\dot{q}_i + \omega_i^2 q_i = (\vec{u}_i^T \vec{f}) \sin(\omega t) \end{cases}$$

$$\text{With damping ratios: } \zeta_i = \frac{1}{2} \left(\frac{\alpha}{\omega_i} + \beta\omega_i \right)$$

4. Independently solve each SDOF modal ODE

$$q_i(t) = q_{ic}(t) + q_{ip}(t)$$

$$q_{ic}(t) = e^{-\zeta_i\omega_i t} (A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t))$$

$$q_{ic}(t) = B_1 \cos(\omega t) + B_2 \sin(\omega t)$$

- NOTE: Three frequencies present $\{\omega, \omega_i, \omega_{id}\}$ (which assumes sys. is underdamped, otherwise $\lambda_i = -\zeta_i\omega_i \pm \omega_i\sqrt{\zeta_i^2 - 1}$)
- Can use any form of harmonic motion eq.
- A_1, A_2 come from fitting to ics
- B_1, B_2 are determined from particular sol.

5. Inverse transform to physical coordinates, and plug ics into **total solution** to solve for remaining undetermined complementary solution coefficients $\vec{x} = U\vec{q}$

General (Non-Proportional) Damping

1. Solve transient/complementary

$$\text{let: } \vec{x}_c(t) = \vec{u}e^{\lambda t} \quad \vec{u} \in \mathbb{C}^n \quad \det(\lambda^2 M + \lambda C + K) = 0$$

2. Solve for eigenvalues and eigenvectors (with MATLAB)

$$[\vec{u}_1 \ \dots \ \vec{u}_n], \text{diag}(\lambda_1 \ \dots \ \lambda_n) = \text{polyeig}(K, C, M)$$

3. Assemble complementary

$$\vec{x}_c(t) = \sum_{i=1}^n (A_i \vec{u}_i e^{\lambda_i t}) + \underbrace{\text{complex conjugate}}_{\text{of sum at left}}$$

A_i are determined by fitting ics to **total response**

4. Solve for the particular solution Solve for $\vec{x}_p(t)$ using the complex form, and take either the Real or Imaginary part of the solution to match to forcing function

$$\text{let: } \vec{y}(t) = \vec{Y}e^{j\omega t} \quad \vec{Y} \in \mathbb{C}^n \quad \omega = \text{driving freq.}$$

$$\vec{x}_p(t) = \begin{cases} \text{Im}\{\vec{y}(t)\} & \vec{f} \sim \sin \\ \text{Re}\{\vec{y}(t)\} & \vec{f} \sim \cos \end{cases}$$

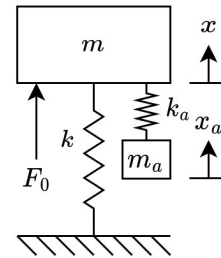
$$\vec{Y} = G(\omega)\vec{f}(t) \quad G(\omega) = [-\omega^2 M + j\omega C + K]^{-1} = \text{FRF gain}$$

$$\vec{Y}(\omega) = \begin{bmatrix} |Y_1|e^{j\phi_1} \\ \vdots \\ |Y_n|e^{j\phi_n} \end{bmatrix} \quad \vec{x}_p(t) = \underbrace{\text{Im}\{\vec{y}(t)\}}_{\text{Re}\{\} \text{ for cos}} = \begin{bmatrix} |Y_1| \sin(\omega t + \phi_1) \\ \vdots \\ |Y_n| \sin(\omega t + \phi_n) \end{bmatrix}$$

$|Y_i|, \phi_i$ are determined while fitting to the particular

5. Plug ics into **total solution** to solve for remaining undetermined complementary solution coefficients A_i

Vibration Absorber



$$\begin{bmatrix} m & 0 \\ 0 & m_a \end{bmatrix} \ddot{\vec{x}} + \begin{bmatrix} k + k_a & -k_a \\ -k_a & k_a \end{bmatrix} \vec{x} = \vec{f} \sin(\omega t)$$

$$\vec{y}_p(t) = \vec{Y}e^{j\omega t} \implies \vec{x}_p(t) = (-\omega^2 M + K)^{-1} \vec{f} \sin(\omega t)$$

$$\vec{x}_p(t) = \frac{F_0 \sin(\omega t)}{\det(K - \omega^2 M)} \begin{bmatrix} k_a - \omega^2 m_a \\ k_a \end{bmatrix}$$

$$(k_a - \omega^2 m_a) \rightarrow 0, x_{pm} \rightarrow 0 \text{ as desired!}$$

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